Software Requirements Specification for Lake: subtitle describing software

Team #10, CinqGarçons
Ethan Walsh
Colin Chambachan
Matthew Nesbitt
Jinwoo Hong
Gary Qin

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Table 1: Revision History

Date	**Developer(s)**	**Change**
Date1	Name(s)	Description of changes
Date2	Name(s)	Description of changes
...	...	...

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Changes from Original SRS Meyer Template

This document follows the requirements template in Meyer's *Handbook of Requirements and Business Analysis* (**?**), which organises a specification into four chapters: Goals (G), Environment (E), System (S), and Project (P). Meyer's template is a book chapter rather than a course template, so we have adapted it. Table 2 records each structural change and the reason for it. The LaTeX structure is adapted from a Meyer-template specification written by a previous capstone team and shared with us by our teaching assistant.

Table 2: Changes from Template

Change	**Reason**
Glossary (E.1) moved to the front of the document	Terms and acronyms must be defined before first use, and information should live in one place. E.1 now points here.
Requirement tables added to each PEGS chapter	Meyer's template describes requirements in prose. Labelled tables of functional (FR) and non-functional (NFR) requirements give each requirement an ID that can be prioritized, traced and verified.
Added G.8 User Profiles	Meyer lists stakeholders (G.7) but has no place for the distinct characteristics of each user group.
Added E.7 Standards and Legal/Regulatory Factors	Meyer's Environment chapter has no explicit home for standards, licensing or legal constraints.
Added S.4 Normal Operation and S.5 Undesired Event Handling	Separates expected behaviour from error handling instead of mixing both into S.2 Functionality.
Added S.9 Formalizations	Provides a place for mathematical or formal specification of the parts of the system that benefit from it.
Added Safety and Security Requirements	Lets the requirements from the Hazard Analysis be ported into this document.
Added Requirement Dependencies	Shows traceability between requirements visually.
Added References	Meyer's template has no references section.
Added Appendices (Generative AI Use Disclosure, Reflection)	Required by the course.
Added S.10 Likely changes to requirements	Required by the SRS rubric
Added S.11 Unlikely changes to requirements	Required by the SRS rubric
Added Appendix (User Interview)	Records the questions put to a potential user and their answers, so the requirements sources in G.7 can be checked.

Glossary

- **Activation** – An intermediate value that is computed during the text generation of an LLM.
- **Device pool** – A collection of nodes that comprise the Lake network.
- **FFI (Foreign Function Interface)** – An interface that allows code written in different languages to call one another.
- **FLOP/s (Floating-point Operations per Second)** – The number of floating point arithmetic operations performed per second. Used as a measure of computational performance.
- **Inference** – The process of using an LLM to generate text.
- **Inference Engine** – The software module that loads the LLM and executes its model layer partitions to generate output.
- **KV cache** – The cache of key-value pairs that are used to store the activations of an LLM.
- **Lake** – A decentralized network of devices that work together to serve LLM inference via model layer partitioning.
- **Lake Client** – The user-facing client interfaces for both consuming the API responses and configuring the Lake system.
- **Latency** – The elapsed time between the start of an inference request and a defined event, such as the completion of the first token, or the completion of the entire response.
- **LLM (large language model)** – An artificial intelligence model that is trained on a large dataset of text and is used to generate text.
- **Model layer partitioning** – The distribution of an LLM's layers across a device pool.
- **Node** – A participating device within the Lake network.
- **Quantization** – The process of converting an LLM's weights to a lower precision format to reduce memory usage.

- **Session** – A unit of interaction between a user and Lake to serve one or more inference requests.
- **Token** – A unit of text that is processed by an LLM, typically a word or a subword.
- **Token throughput** – The number of tokens that can be generated by an LLM per unit time.
- **Weights** – The parameters of an LLM that determine its behavior when generating text.

(G) Goals

This following explains why Lake exists and what it is for, in terms a reader outside the team can follow. How the system is built is left to the System chapter (Section E.7), and the world it runs in to the Environment chapter (Section G.8.0.3).

G.1 Context and Overall Objective

LLMs have become an everyday tool, and almost everyone who uses one does so through a commercial provider's web interface or application programming interface (API). The alternative is to run an open model on your own machine, which keeps your data at home but limits you to whatever fits in one device's memory. Most households, offices, and labs already own several computers, and most of them sit idle most of the time.

Lake runs a single pre-trained LLM across several consumer devices on one local network. The model is split by layer, each device holds and runs only its share, and the devices pass intermediate results between them so that together they answer a prompt as if the model were running on one machine. The objective is to let people run models that are too large for any one of their devices, privately and on hardware they already own, and to keep the model running as devices join and leave the network.

Figure 1 shows where Lake's boundary sits. Lake is the software on each participating device. The devices themselves, the network between them, the model file, and the tools a user sends prompts from are all outside the boundary.

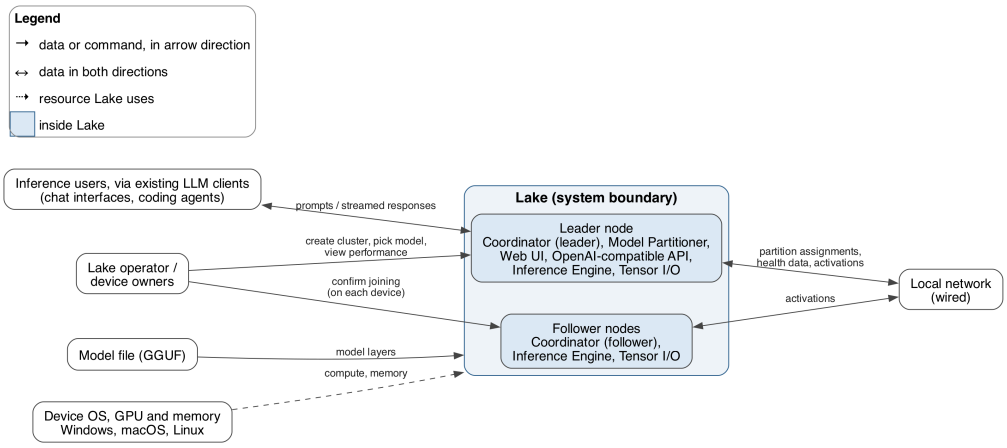


Figure 1: Context diagram showing Lake's boundary and the people, tools, files, and hardware around it.

G.2 Current Situation

Today there are two ways to use an LLM. The first is a hosted service from one of the major inference providers. These give the best model quality, but every prompt leaves the premises, the cost grows with use, and the user has no control over how their data is handled. For a clinic or a law office this is often not acceptable at all.

The second is local inference with a tool such as llama.cpp (**?**) or Ollama (**?**). The model runs in one process on one machine, so the largest model a user can run is set by the memory of their single best device. A laptop that runs a small model well may run a mid-sized one slowly and a large one not at all, even when three other computers sit on the same desk.

A few projects already spread a model over several machines: exo (**?**), prima.cpp (**?**), and the remote procedure call (RPC) backend of llama.cpp. They assign layers to devices once, when the model loads, and the assignment stays fixed for the session. If a device drops off the network the session fails, and a device added later goes unused until everything is restarted. Lake builds on the same idea but makes the assignment something the system can change while it is running. Table 3 summarises how Lake compares. Mixed operating systems are marked "partly" for the existing tools because exo runs on Apple silicon through MLX and only on the CPU on Linux, with no

Windows build (**?**), and prima.cpp supports Linux and macOS (**?**); only the llama.cpp RPC backend runs on all three.

Table 3: Lake compared with the current options

	Hosted service	**Single-device local**	**Existing distributed tools**	**Lake**
Data stays on the user's network	No	Yes	Yes	Yes
Cost per use	Per token	None	None	None
Largest model	Provider's choice	One device's memory	Pooled memory	Pooled memory
Survives a device leaving	N/A	N/A	No, session fails	Yes, layers reassigned
Uses a device that joins later	N/A	N/A	No, needs restart	Yes
Mixed operating systems in one pool	N/A	N/A	Partly	Windows, macOS, Linux

G.3 Expected Benefits

The benefits below follow the goals in the Problem Statement: the four numbered goals and concurrent inference, which is a goal in its own right because serving several requests at once is one of the main advantages of splitting a model by layers. Figure 2 is a goal model thats reflects this. Each benefit below states the measure we will use to check it, and the requirements that deliver them are in Section S.2.

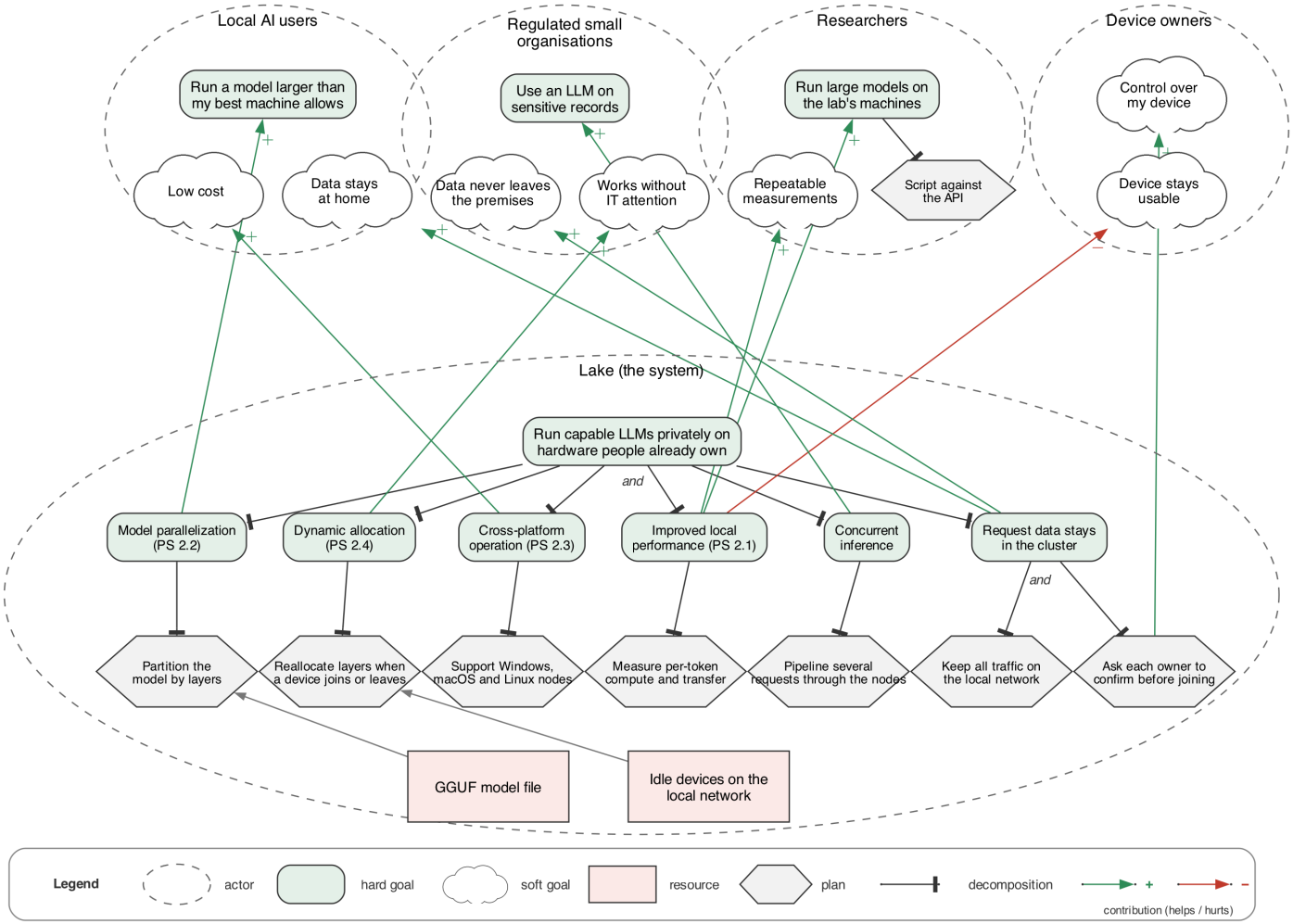


Figure 2: Goal model for Lake and its stakeholders.

Larger models on existing hardware

A user can run a model that does not fit on any one of their devices. The measure is the largest model, by parameter count, precision, and context window, that a cluster can serve compared with the largest the single best device in that cluster can serve alone. This is the primary benefit, and the Model Size requirement in Section S.2 states the target.

Privacy

Prompts, responses, and the model's intermediate data stay on the devices in the cluster and never go to an external provider. The measure is that no request data leaves the cluster during a session. What privacy does and does not cover in the first release is stated in Section G.6.

Lower cost

There are no per-token or subscription fees, and no new hardware is needed beyond what the user already owns. The measure is that a cluster can be formed entirely from machines the user had before installing Lake.

Resilience to devices coming and going

When a device leaves or joins, Lake reassigns layers and keeps serving requests as long as the remaining cluster has enough memory. The measures are whether requests are served after a reassignment and how long the reassignment takes, which Section S.9 defines precisely.

Mixed platforms in one cluster

Windows, macOS, and Linux machines can be combined, so a user is not limited to devices of one type. The measure is a working cluster that mixes all three.

Works with the tools people already use

Lake exposes the same API format as the major hosted providers, so existing chat interfaces and coding assistants can point at Lake instead of a cloud service with no changes on their side. The measure is an unmodified third-party client completing a session against Lake.

Several users at once

While one request's token is being computed on one node, another request's token can be computed on the next, so a cluster can serve several users without each one waiting for the others. The measure is the number of requests the cluster can serve at once before any single request slows by more than the tolerance set in the Scalability requirement in Section S.2, and how that number grows with the number of nodes.

Speed, where the split allows it

Response time and throughput (tokens per second) for a single request are secondary measures. A split model is not expected to beat a single device on a model that already fits on that device, since every token still passes through every layer in order. The gain is in capacity, and in throughput when several requests are served at once. The practitioner we interviewed (see Section G.7.2) expected the same: a split over an office network would likely be slower than one good GPU for live chat, but worth it for background work such as re-indexing documents, OCR, and overnight test runs, which is a use we had not considered.

G.4 Functionality Overview

This section is a capsule version of the System chapter. Each high-level requirement here maps to one or more of the components in Section S.1 (the Coordinator, the Model Partitioner, the Inference Engine, Tensor I/O, and the Web UI) and to detailed requirements in Section S.2.

G.4.1 Functional Requirements

These are the high-level requirements, stated for the reader of the Goals chapter. Each ends with the detailed, verifiable requirements in Section S.2 and the Environmental Requirements table that realise it, so a change in either place can be traced to the other.

- **FR.G.4.1 Form a cluster with consent.** Lake runs on each device. From the web user interface (UI) on one device, which becomes the leader node, the operator sees the other devices on the local network that are running Lake and chooses which to include. Each chosen

device asks its owner to confirm before it joins, so no device takes part without consent. *Realised by* FR.S.1.4, FR.S.5.1, FR.S.5.2, FR.S.1.17, FR.S.1.18, FR.S.1.20, FR.S.1.21, FR.S.5.7, and NFR.S.12.

- **FR.G.4.2 Load and partition a model.** The operator picks a model in GGUF format (**?**), the file format used by llama.cpp. Lake measures each node's memory, compute, and network performance, decides which range of layers each node will hold, and has each node load only that range. *Realised by* FR.S.5.3, FR.S.1.8, FR.S.2.1 to FR.S.2.4, FR.S.1.6, FR.S.1.19, and FR.S.3.1.

- **FR.G.4.3 Run inference across the cluster.** A prompt enters at the first node, the activations pass from node to node in layer order, and generated tokens stream back to the user one at a time. Each node keeps the key-value (KV) cache for its own layers, so nothing about the conversation has to be re-sent between tokens. Several requests can be in flight at once, each at a different node, and their state is kept apart. *Realised by* FR.S.3.2, FR.S.3.3, FR.S.4.1, FR.S.4.3, FR.S.4.4, and, for concurrent requests, FR.S.1.14, FR.S.3.5, FR.S.4.6, and NFR.S.5.

- **FR.G.4.4 Reallocate when the cluster changes.** When a node leaves, fails, or joins, the leader re-partitions the model across the nodes that remain and resumes serving, retrying any request that was in progress. If the cluster no longer has enough memory for the model, the user is told why rather than left with a hung session. *Realised by* FR.S.1.10 to FR.S.1.13, FR.S.1.22, FR.S.2.2, and FR.S.5.4.

- **FR.G.4.5 Measure performance.** Lake records compute time on each node and transfer time between nodes. These measurements drive the partitioner and are shown in the web UI, which highlights the node or link that is the bottleneck and compares the cluster against the single best machine in it, so the operator can see whether pooling is paying off. *Realised by* FR.S.1.2, FR.S.1.3, FR.S.1.7, FR.S.1.23, FR.S.3.4, FR.S.4.5, FR.S.5.5, FR.S.5.6, and NFR.S.1.

- **FR.G.4.6 Serve prompts through a standard API.** The leader node exposes an HTTP API in the OpenAI format (**?**), including streaming and tool calling, so any client that can talk to a hosted provider can talk to Lake. *Realised by* FR.S.1.9 and NFR.E.3.

Two of these carry most of the value: FR.G.4.2 and FR.G.4.4. Partitioning is what lets a model larger than any one device run at all, which is the primary goal of the project. Reallocation is what sets Lake apart from exo and the llama.cpp RPC backend, both of which fix the split when the model loads. Without the first there is no product; without the second there is nothing a user could not already get elsewhere.

G.4.2 Non-Functional Requirements

These state how well Lake must do what Section G.4.1 describes. Each is a quality the system must have, not a design choice, and each names the detailed requirement that gives it a measurable target.

- **NFR.G.4.1 Request data stays in the cluster.** No prompt, response, or intermediate data is sent to any device outside the cluster, none is kept after a request completes, and Lake sends no telemetry. This is the privacy benefit, and it is the reason an organisation handling export-controlled or medical data could use Lake at all. *Realised by* NFR.S.10, NFR.S.11, NFR.S.12, and NFR.E.5.

- **NFR.G.4.2 Runs on Windows, macOS, and Linux.** Each is a supported platform for any node, and one cluster can mix all three. This is the cross-platform goal from the Problem Statement. *Realised by* NFR.S.3, NFR.S.13 to NFR.S.15, and NFR.E.1.

- **NFR.G.4.3 The split does not change the model.** A model run through Lake produces output of the same quality as the same model run on one machine, within the tolerance set by NFR.S.6. Users would not trust a system that made their model worse. *Realised by* NFR.S.6, FR.S.2.3, and NFR.E.4.

- **NFR.G.4.4 License compliance.** Lake respects the licenses of the software it builds on and of the models it runs, as set out in Section E.7. *Realised by* NFR.E.7 and NFR.E.8.

- **NFR.G.4.5 Installable without expert help.** A user can install Lake on each supported platform from an installer or a single package-manager command, without building from source or editing configuration files. Both the home user and the organisation in Section G.8 said

they would not deploy a tool that needs more than that. *Realised by* NFR.S.3.

- **NFR.G.4.6 Serves more users as nodes are added.** The number of requests a cluster can serve at once without any one of them slowing beyond a set tolerance grows with the number of nodes. This is the measure behind the concurrent inference goal. *Realised by* NFR.S.5.

Privacy and fidelity matter most. A local inference tool that leaked data or degraded the model would fail the two groups it is meant for: the privacy-bound organisation and the user who already has a model they like.

G.4.3 RACI Matrix

Table 4 shows who is responsible, accountable, consulted, and informed for each high-level requirement, using the stakeholders from Section G.7.

Table 4: RACI matrix for the high-level requirements

Requirement	Local AI users	Regulated small orgs	Small AI labs	Device owners	Development team	Instructor and TA	Society
FR.G.4.1 Form a cluster with consent	C	C	C	C	R, A	I	–
FR.G.4.2 Load and partition a model	C	I	C	I	R, A	I	–
FR.G.4.3 Run inference across the cluster	C	I	C	I	R, A	I	–
FR.G.4.4 Reallocate when the cluster changes	C	C	C	C	R, A	I	–
FR.G.4.5 Measure performance	C	I	C	I	R, A	I	–
FR.G.4.6 Serve prompts through a standard API	C	I	C	–	R, A	I	–
NFR.G.4.1 Request data stays in the cluster	C	C	C	C	R, A	I	I
NFR.G.4.2 Runs on Windows, macOS, and Linux	C	C	C	C	R, A	I	–
NFR.G.4.3 The split does not change the model	C	I	C	–	R, A	I	I
NFR.G.4.4 License compliance	I	C	I	I	R, A	C	I
NFR.G.4.5 Installable without expert help	C	C	C	C	R, A	I	–
NFR.G.4.6 Serves more users as nodes are added	I	C	C	I	R, A	I	–

Table 5: RACI legend

Symbol	Definition
R	Responsible: who does the work
A	Accountable: who answers for it being done
C	Consulted: who gives input
I	Informed: who is kept up to date
–	No involvement

G.5 High-Level Usage Scenarios

These are the main ways people use Lake, stated in user terms without system detail or error cases. Figure 3 shows them as use cases with the three actors from Section G.8. Each use case is expanded, with special and erroneous cases, in Section S.6.

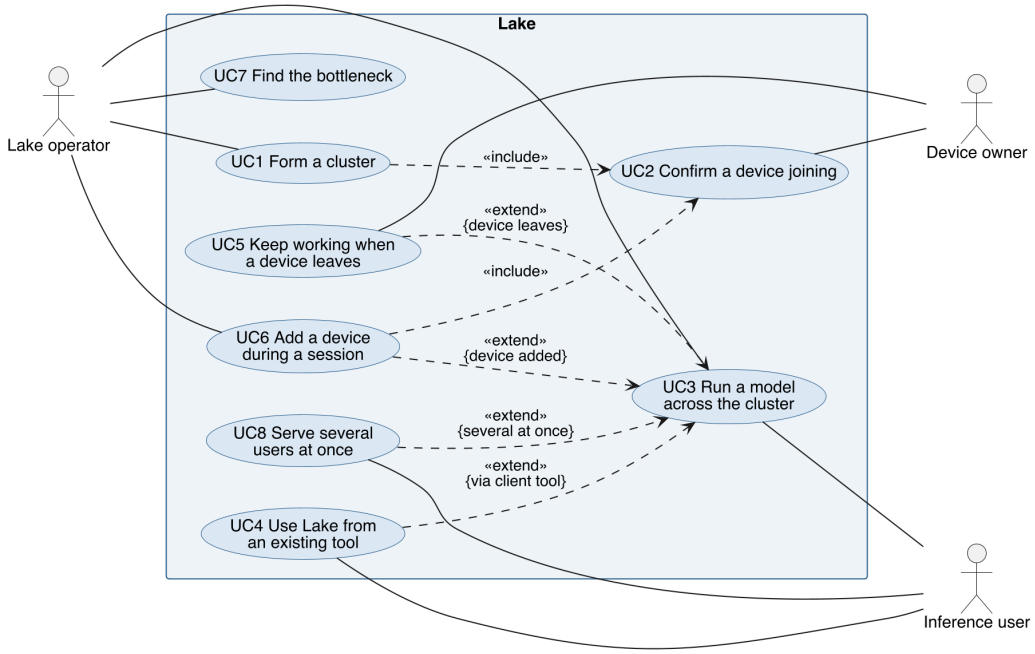


Figure 3: High-level use case diagram. Stick figures are actors, ellipses are use cases, and the box is Lake. A solid line joins an actor to a use case they take part in. A dashed arrow labelled *include* means the source use case always performs the target; one labelled *extend* means the source adds to the target under the condition in braces.

1. **UC1 Form a cluster**

   - The operator starts Lake on each computer they want to use.
   - On one of them, the operator opens Lake and sees the other computers on the network that are running it.
   - The operator selects the computers to include.

- Each selected computer's owner confirms (UC2).
- The cluster is shown with each computer's memory and platform.

2. **UC2 Confirm a device joining**
   - A prompt appears on the device asking whether it may join the named cluster.
   - The owner accepts or declines.
   - If accepted, the device joins and the owner can see what it is doing; if declined, it is left out.
3. **UC3 Run a model across the cluster**
   - The operator chooses a model file, which may be larger than any one computer can hold.
   - Lake reports that the model is loaded across the cluster.
   - A user sends a prompt.
   - The response streams back word by word.
   - The user sends further prompts in the same session.
4. **UC4 Use Lake from an existing tool**
   - A user opens the chat interface or coding assistant they already use.
   - They point it at Lake's address instead of a cloud provider.
   - They use the tool as before; prompts and responses go through Lake and stay on the local network.
5. **UC5 Keep working when a device leaves**
   - During a session, a computer in the cluster is closed or disconnected by its owner.
   - Lake tells the operator which device left and that it is reorganising.
   - The user's current prompt is answered, or re-run, without being resubmitted. A partial answer is never presented as a complete one.

- If the remaining computers cannot hold the model, the operator is told why and what would fix it.

6. **UC6 Add a device during a session**

   - The operator selects a new computer that has appeared on the network.
   - Its owner confirms (UC2).
   - Lake reorganises to use the new computer and shows the new arrangement.
   - The user continues prompting and sees the change in speed.

7. **UC7 Find the bottleneck**

   - Responses feel slow, so the operator opens the cluster view.
   - The view shows how long each computer and each connection takes per word generated, and marks the slowest.
   - The operator decides whether to swap a cable, remove a computer, or add one.

8. **UC8 Serve several users at once**

   - Two or more users send prompts to the same cluster at the same time, from their own tools.
   - Each sees their own response stream back, and none sees another's prompt or answer.
   - Each response arrives about as fast as it would have alone, until the cluster is busy enough that the operator sees requests queue in the cluster view.

G.6 Limitations and Exclusions

Lake is deliberately narrow. The following are out of scope. The stretch goals from the Problem Statement are listed as exclusions until the team decides to take them on. The likely and unlikely changes listed at the end of the System chapter record which of these may change.

- *No training or fine-tuning.* Lake runs pre-trained transformer models that ggml (**?**) supports. Lake does not train, fine-tune, quantize, or otherwise modify a model, and does not change the quality of its output.

- *Trusted network and devices.* In the first release, traffic between nodes is not encrypted and devices are not authenticated beyond the owner's confirmation. Privacy therefore means that data never leaves the user's own network, not that it is protected from other devices on that network. Users must run Lake only on devices they trust and a network they control. Encryption and authentication are a stretch goal.

- *Supported hardware.* Each device needs a GPU from Nvidia, AMD, or Apple, because inference on a CPU alone is too slow to be useful in a cluster. Smartphones are a stretch goal.

- *Wired connections first.* The first revision targets wired Ethernet. Wireless operation is a stretch goal because its latency and variability may make the split model too slow.

- *Not faster for models that already fit.* Lake does not speed up a model that already runs well on one device. The benefit is running models that would not otherwise fit.

- *No model hosting.* Lake does not download or distribute model weights. Users supply a model file under that model's own license.

G.7 Stakeholders and Requirements Sources

G.7.1 Stakeholder Definition

Table 6 lists the categories of people who can affect Lake or be affected by it. Direct stakeholders use or build the system; indirect ones are affected by it without touching it. The three direct user groups are developed into personas in Section G.8.

Table 6: Stakeholders

Stakeholder	**Type**	**Description**	**Justification**
Local AI users	Direct	People who already run models on their own machines with tools like llama.cpp or Ollama and want to run larger ones.	The primary hands-on users, and the group the team itself belongs to. Their habits set the baseline Lake has to beat.
Organisations in regulated industries	Direct	An engineering firm with export-controlled data, a medical clinic, or a law office that wants AI assistance on sensitive records but cannot let the data leave the building. The one we interviewed already runs an in-house assistant for about two hundred staff on its own GPUs.	For them privacy is a legal obligation, not a preference. They are the clearest case for running inference on premises, the reason the privacy benefit has to be exact about what it covers, and the source of the deployment needs in Section G.8: managed machines only, no telemetry, and an off switch.
Small AI labs and research groups	Direct	Groups with a few machines and a modest budget who want to run large models without renting cloud GPUs.	They push the model-size and measurement requirements hardest and are the most likely to use the API directly.

Table 6 continued

Stakeholder	Type	Description	Justification
Device owners	Direct	People whose computer joins a cluster, who may not be the person running it.	Their device gives up memory, compute, battery, and bandwidth while it takes part. They must consent, be able to see what their device is doing, and be able to pause it; the interviewee's first condition for pooling office machines was that Lake back off the moment the owner is working.
Development team	Direct	The five students building Lake as a capstone project.	Responsible for every requirement here and bound by the course timeline, which limits scope.
Course instructor and teaching assistant	Indirect	Assess the project against the course rubrics and give feedback on each deliverable.	Their feedback changes the requirements, and the deliverable schedule in Section P.3 is theirs.
Open-source projects Lake builds on	Indirect	ggml, llama.cpp, and the GGUF format, with their maintainers and licenses.	Their interfaces decide which models and hardware Lake can support, and their licenses constrain how Lake is distributed.

Table 6 continued

Stakeholder	Type	Description	Justification
Society	Indirect	The public, and in particular people and organisations who cannot pay for hosted inference or a high-end machine.	Lake lowers the cost of using capable models privately, which broadens who can use them. Health data is the clearest case. The same applies outside wealthy, English-speaking markets: a school or small business anywhere with a few ordinary computers can run an open model in its own language. Lake also raises energy use on participating devices, and a model run through Lake carries whatever biases and risks it had before, since Lake does not alter it. The Hazard Analysis covers the safety side.

G.7.2 Requirements Sources

Beyond the stakeholders above, the requirements in this document come from the following sources.

G.7.2.1 Project documents. The Problem Statement and Development Plan set the goals and their metrics, the stretch goals, the hardware

limits, and the technology direction. The Hazard Analysis contributes the safety and security requirements and the critical assumptions.

G.7.2.2 Peer review. Reviews of the Problem Statement from Team 07 and from a reviewer on another team led to a clearer statement of what privacy covers in the first release and to a measurable definition of dynamic allocation. Both are tracked as issues in the project repository.

G.7.2.3 Teaching assistant. Meetings with the teaching assistant recommended this template and shaped how the requirements are organised.

G.7.2.4 User interview. On October 16, 2026 we sent the eight questions in Appendix 1 to an AI developer at an aerospace engineering firm, who answered in writing. He builds and runs the company's in-house assistant for about two hundred staff, entirely on two GPU machines, because much of the firm's data is export-controlled and cannot go to a cloud provider. The interview confirmed the privacy benefit as the reason such organisations run locally at all, confirmed the OpenAI-compatible API as a must (with streaming, tool calling, and structured output), and confirmed the bottleneck view in UC7. It corrected us in three places: a split over an office network is expected to be slower than one good GPU for live chat, so the honest pitch is capacity and background work rather than speed; a retried request must never repeat side effects or present a cut-off answer as complete (UC5); and a deployment in an organisation needs managed machines only, no telemetry, no automatic updates, and a way for each owner to pause their machine, which shaped the second persona and the device-owner stakeholder.

G.7.2.5 Regulation and guidance. No law requires an organisation to run its models locally, but several make it the simplest way to comply. Canada's private-sector privacy law, PIPEDA (**?**), keeps an organisation accountable for personal information it passes to a third party. Export control regimes such as Canada's Controlled Goods Program under the Defence Production Act (**??**) restrict who may examine or possess controlled technical data, which includes software and documentation, and this is the kind of rule that keeps the interviewee's firm off hosted services. NIST's generative AI profile (**?**) lists data privacy, information integrity, and information security among the risks an operator of a generative model is expected to manage.

These documents are the background for NFR.G.4.1 and the no-telemetry expectation in Section G.8. Which of them bind a given deployment depends on who deploys Lake; the standards and licences that bind Lake itself are in Section E.7.

G.7.2.6 The team as users. The team's own experience running models locally is the source of the first persona and of the use cases in Section G.5.

G.7.2.7 Existing distributed inference tools. exo (**?**), prima.cpp (**?**), and the RPC backend of llama.cpp (**?**) were studied to learn what current systems do and where they fall short. The gap they share, a split that is fixed when the model loads, is the origin of the dynamic allocation goal.

G.7.2.8 Libraries and formats. The ggml and llama.cpp documentation and source (**???**) define which models and hardware can be supported and how a model can be split. The OpenAI API reference (**?**) defines the request and response format Lake's API follows.

G.7.2.9 Course material and the template. The course templates, rubrics, and lectures, and Meyer's handbook (**?**), which this document's structure follows.

G.8 User Profiles

The Environment chapter distinguishes three roles: the inference user who sends prompts, the Lake operator who sets up and runs the cluster, and the device owner who contributes a machine. The personas below represent the three direct user groups from Table 6 and show how those roles land on real people. They are composites, not individuals.

G.8.0.1 Local AI user *Priya Nair, 29, software developer*
"I've got a desktop with a good GPU and two laptops that mostly sit there. I want to run the 70B model, not the 8B one."

Priya runs models on her gaming desktop with Ollama and llama.cpp and knows what quantization and context length mean. She reads release notes, has opinions about which models are good, and is comfortable with a command line, installing software on several operating systems, and reading

logs. Her problem is that the model she wants does not fit on her desktop, and a bigger GPU costs more than she will spend on a hobby. She is the operator, the device owner, and the inference user all at once. She will tolerate rough edges if the system works, which makes her the right first user, and she drives the requirements for straightforward installation on each platform, GGUF model support, and clear reporting of what each device loaded.

G.8.0.2 Operator in a regulated organisation

Nadia Reyes, 34, AI developer at an engineering firm

"We removed the cloud provider from our pipeline. Export-controlled documents can't leave the building, so everything runs on our own GPUs."

Nadia is based on the practitioner we interviewed. She builds and runs an in-house assistant used by two hundred colleagues, serving a 27-billion parameter model from two workstation GPUs, and she has already tuned prompt caching and speculative decoding to get the speed she needs. She knows inference well, reads vLLM metrics daily, and tests every release before it reaches users. She is the operator; her colleagues are the inference users through a chat interface and office add-ins; the company owns the devices. What stops her running a larger model is memory per card, and her interest in Lake is pooling idle workstations for background jobs such as re-indexing, OCR, and overnight evaluation, not live chat. She would deploy it only with signed installers pushed by IT, no telemetry, no automatic updates, company-managed machines only, and a pause switch on every node. She drives the requirements for the OpenAI-compatible API with streaming and tool calling, for comparing the cluster against the single best machine, for exportable metrics, for the no-telemetry privacy guarantee, and for retries that never repeat side effects.

G.8.0.3 Researcher

Dr. Mei Lin, 35, postdoctoral researcher leading a four-person NLP group

"We have six workstations and no cloud budget. I need numbers I can put in a paper."

Mei understands transformer models and inference and wants to compare models and settings. She will use the API directly and script against it, and she cares about repeatable measurements of speed and overhead as much as about the model running at all. She is usually both the operator and

an inference user, and her lab mates are the device owners. She drives the requirements for the OpenAI-format API, the per-token performance measurements, and for comparing a split run against a single-device run.

(E) Environment

E.1 Glossary

The glossary has been moved to the front of the document (see the Glossary and the Changes from Original SRS Meyer Template).

E.2 Environment Components

- **Inference users (a.k.a users)** – Individuals who interact with the Lake network to serve inference requests.
- **Lake operators** – Individuals who manage the Lake network, including the device pool and the model.
- **Device owners** – Individuals who contribute one or more nodes towards the device pool.
- **Operating systems and drivers** – Windows, macOS, and Linux operating systems that have the supported drivers to facilitate inference requests. Android and iOS operating system support is a stretch goal.
- **Network equipment** – Routers, switches, Thunderbolt cables, and any other hardware necessary to connect the nodes to the network. Connectivity via Wi-Fi is a stretch goal.
- **Model** – Pre-trained and open-source model weights that are permitted to be downloaded and utilized by the Lake network via model layer partitioning.
- **External API consumers** – Other services that utilize the Lake API to serve inference requests. This acts as the main interface for consumers to use Lake, such as coding agent harnesses (e.g. Claude Code, Codex, Pi).

E.3 Constraints

- **Device hardware** – Each node should have a sufficient CPU, GPU, memory, and storage such that its assigned model layers, KV cache, and other Lake software components can fit and run within the device,

such that distributed inference has both a higher token throughput and lower latency than an equivalent single-device inference.

- **Network communication** – Network communication, whether wired or wireless, should have a sufficiently high bandwidth and low latency. Inter-node communication overhead may make distributed inference slower than single-device inference.
- **Platform compatibility** – Nodes within the Lake network must operate on Windows, macOS, and Linux (as well as Android and iOS as stretch goals) as hardware acceleration and inference platforms are limited by the available drivers on these operating systems.
- **API compatibility** – Lake must implement a standardized OpenAI API spec such that existing external clients can easily utilize Lake, (**?**), as specified in Section S.3.
- **Model architecture** – Only supported LLM architectures that Lake's model layer partitioning supports can be used.

E.4 Assumptions

- **Device owners authorize their nodes for use** – Device owners have authorized their device to be used in the Lake network as a node for serving inference.
- **Participating nodes are trusted devices within the Lake network** – Each node is trusted to run and process potentially sensitive inference data.
- **Permitted model use** – Lake operators are using permitted models for inference that adhere to the respective licensing.
- **Network reachability** – Nodes within the Lake network must be able to communicate over a network connection.
- **Model suitability** – Models that are suited for a given task are used when Lake is serving inference (e.g. a larger model for logic/coding-based tasks, a smaller model for conversational tasks).
- **Trusted local network** – The local network connecting the participating nodes is private and trusted by the operator.

E.5 Effects

- **Resource usage** – Device owners' CPU/GPU, memory, and disk usage will increase while the device is part of the Lake network, potentially making the device too resource constrained to multi-task or perform other activities. When not in use, any cached data or the stored layers can continue to utilize disk space.
- **Energy usage** – Devices will consume more power, increasing energy usage, heat generated, fan usage, and decreasing battery life if unplugged.
- **Network bandwidth usage** – The Lake network may utilize a large amount of a network's bandwidth, inhibiting other devices or processes from accessing the network.
- **Logged data** – During inference, logs and other diagnostic information can be stored and persist on the node's disk. Logs must exclude sensitive request data as specified in SR5-2.
- **Access to inference** – Users can gain access to inference that they previously would not be able to get within a single device or with a paid service.

E.6 Invariants

- The provided model must remain unchanged. Loading the model and partitioning its layers must not mutate the underlying model weights.
- Inference must be contained within the Lake network, and must not be sent to 3rd-party services.
- All nodes within the Lake network must be using compatible layer partitions that originate from the same model's layers.
- All nodes must serve inference in a way that the activations are performed in the same order as in single-device inference.
- Intermediate data from one request must not be used for another independent request.

E.7 Standards and Legal/Regulatory Factors

Standards:

- **API compatibility** – Lake's inference API will follow the OpenAI API endpoints for requests and response formats, as specified in S.3.
- **Coding conventions** – Black code style and Ruff lint checks will be used for Python components (**??**). Rustfmt and Clippy will be used for Rust components (**??**). Prettier and ESLint will be used for TypeScript components (**??**).

Legal and Regulatory Factors:

- **Software licensing** – Lake is available under the GNU GPL v3.0 license (**?**). All dependencies used in Lake must be compliant and compatible with Lake's intended use and distribution.
- **Model licensing** – Pretrained LLMs used for inference are subject to their respective licenses. Lake must be compliant with these licenses for its use and distribution.

Environmental Requirements

Table 7: Environmental Requirements

Label	**Requirement**
NFR.E.1	The system shall be installable and usable on the supported platforms of Windows, macOS, and Linux, and Android/iOS as a stretch goal. Related to NFR.S.3 and NFR.S.13.
NFR.E.2	The system shall perform model layer partitioning to distribute model inference across various supported devices. Related to SR1-1 and SR1-2.
NFR.E.3	The system shall follow the OpenAI API specification for its request and response API service. Related to FR.S.1.9.
NFR.E.4	The system shall not modify the underlying model weights during model layer partitioning.
NFR.E.5	The system shall contain all inference requests within the Lake network and not send any requests to 3rd-party services. Related to NFR.S.11.
NFR.E.6	The system shall adhere to all codebase formatting and linting conventions.
NFR.E.7	The system shall follow the GNU GPL v3.0 license in applicable use cases and distributions.
NFR.E.8	The system shall adhere to a pretrained model's licensing requirements for use and distribution within Lake.

(S) System

S.1 Components

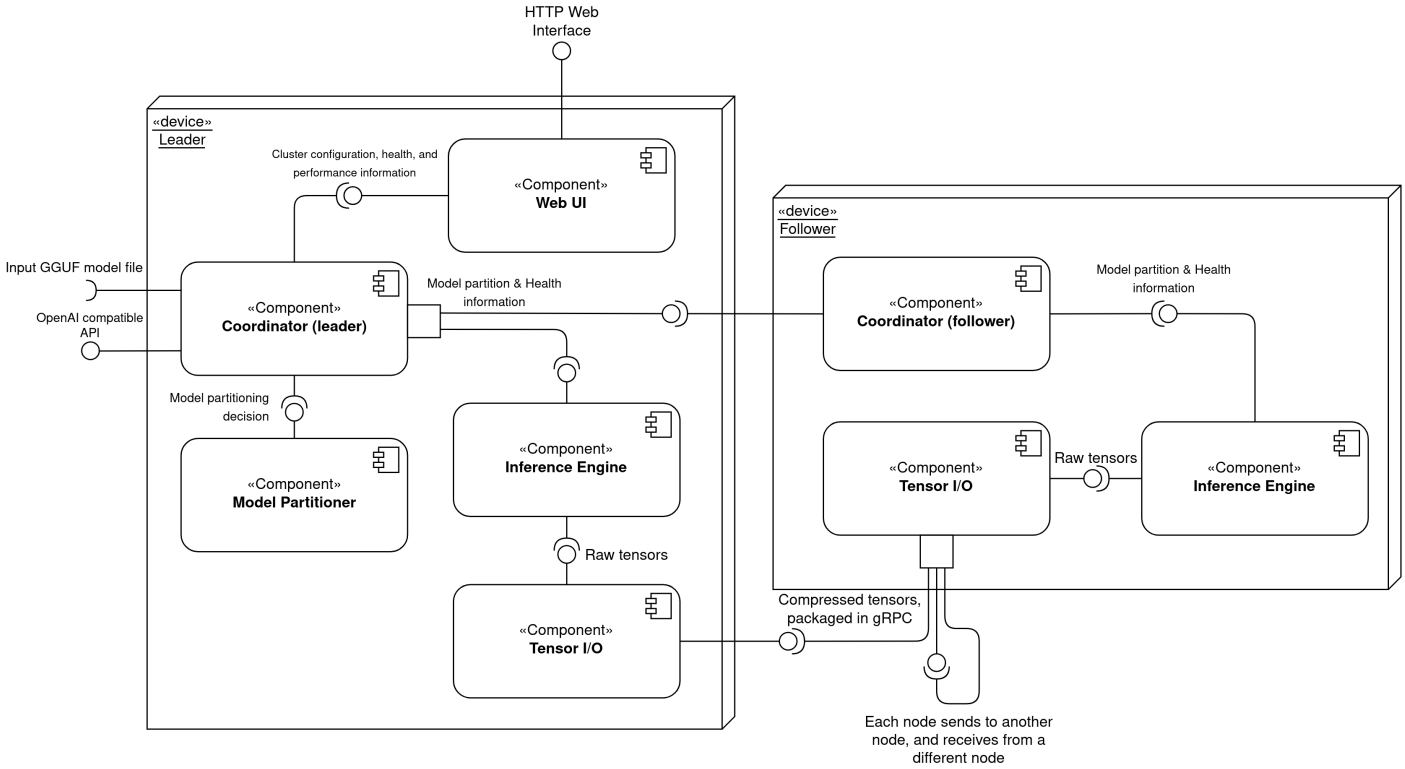


Figure 4: Component diagram.

- **Coordinator** – This component exists on all nodes, but the following responsibilities are only run on the leader node:
  - Providing an OpenAI-API compatible API to the user for use with tools like OpenCode.
  - Tracking cluster health, and detecting node failures.
  - Notifying nodes of partitioning changes.
  - Rebuilding the cluster and saving in-progress requests in the case of node failure, removal, or addition.

– Measuring node connection performance.

On a follower node, it is primarily responsible for managing the inference engine on its device, following the instructions of the leader node.

- **Model Partitioner** – This component determines how to partition the model among the participating devices. It decides at which points to cut the graph, how large each partition is, and which device gets each partition. This component is integral to ensuring that each device is assigned an appropriate partition given the device's computational and networking performance.
- **Inference Engine** – This component is the high-performance inference engine running on each device. It will handle execution of the given model partition for each device. It will call into ggml using the Rust-to-C FFI to control the execution of a device's model partition.
- **Tensor I/O** – This component handles the compressing, sending, receiving, and decompressing of tensors between devices. It will leverage gRPC for high-performance communication, with the potential to use a custom solution should gRPC fail to meet our requirements in practice.
- **Web UI** – This component handles displaying cluster information and structure to the user, as well as allowing user inputs to configure a new cluster, or modify an existing cluster.

S.2 Functionality

S.2.1 Coordinator

Functional Requirements

- **FR.S.1.1** The system shall provide the local Inference Engine with the model partition assigned to the local device.
- **FR.S.1.2** The system shall collect device health/performance information[1] from the local Inference Engine on a 5 second interval. Partially satisfies hazards H0-1 and H0-2 from the Hazard Analysis.

[1] e.g. token throughput, end-to-end partition latency, backpressure.

- **FR.S.1.3** The system shall collect the network health/performance information[2] for established connections from the Tensor I/O component on a 5 second interval. Partially satisfies hazards H0-1 and H0-2 from the Hazard Analysis.
- **FR.S.1.4** The system shall continuously scan for all other devices on the same network running Lake.
- **FR.S.1.5** The system shall provide the upstream and downstream node info to the local Tensor I/O.
- **When running on a leader node:**
  - **FR.S.1.6** The system shall send model partitions to other Coordinators on their respective devices as determined by the Model Partitioner. Partially satisfies hazard H0-2 from the Hazard Analysis.
  - **FR.S.1.7** The system shall monitor cluster performance by gathering performance data from each follower node's Coordinator. Partially satisfies hazards H0-1 and H0-2 from the Hazard Analysis.
  - **FR.S.1.8** The system shall provide the Model Partitioner with the available nodes, and gathered node performance data, to help inform optimal model cut points.
  - **FR.S.1.9** The system shall provide an HTTP API matching OpenAI's API.
  - **FR.S.1.10** The system shall detect failing follower nodes[3]. Partially satisfies hazards H2-1, H4-2 and H4-1 from the Hazard Analysis.
  - **FR.S.1.11** The system shall rebuild the cluster on a node failing or leaving. Partially satisfies hazards H2-1, H4-2 and H4-1 from the Hazard Analysis.
  - **FR.S.1.12** The system shall rebuild the cluster on a node joining.

[2] e.g. RTT, throughput, backpressure.

[3] A "failing" device is defined as severely degraded ($< 50\%$ of historical) token throughput, whether it be due to network or device resource instability.

- **FR.S.1.13** The system shall retry in-progress requests after a cluster rebuild.
- **FR.S.1.14** The system shall accept multiple concurrent requests through the HTTP API. Related to NFR.S.5.
- **FR.S.1.15** The system shall communicate the cluster graph structure to each follower.
- **FR.S.1.16** The system shall provide the Web UI with cluster information, and accept cluster configuration instructions from the Web UI.
- **FR.S.1.17** The system shall not begin inference until all selected followers have confirmed participation. Satisfies NFR.S.12.
- **FR.S.1.18** The system shall only include selected followers in a new cluster. Satisfies NFR.S.12.

- **When running on a follower node:**
  - **FR.S.1.19** The system shall accept model partition assignments from the leader node Coordinator.
  - **FR.S.1.20** The system shall, upon receiving an invite to join a cluster, request that the user confirm or decline participation. Satisfies NFR.S.12.
  - **FR.S.1.21** The system shall not partake in a cluster without user confirmation. Satisfies NFR.S.12.
  - **FR.S.1.22** The system shall not require confirmation to re-join a rebuilding cluster led by the same device, so long as the local device was already participating in the original cluster.
  - **FR.S.1.23** The system shall report collected health/performance data to the leader node Coordinator on a 5 second interval. Partially satisfies hazards H0-1 and H0-2 from the Hazard Analysis.
  - **FR.S.1.24** The system shall assume the leader node has failed if there has been no health query from the leader node for more than 60 consecutive seconds.
  - **FR.S.1.25** The system shall initiate the shutdown of the local node, if the leader node is assumed to have failed. Partially satisfies hazards H4-2 and H4-1 from the Hazard Analysis.

Rationale: The Leader node is the device the user is physically using to interact with the cluster, so failure of that device will remove the user's ability to use the cluster, whether or not the cluster survives. The architecture of Lake will allow the ability to recover from a leader failure to be implemented at a later date.

S.2.2 Model Partitioner

Functional Requirements

- **FR.S.2.1** The system shall assign all devices a partition smaller than their available GPU-accessible memory, with additional room left for runtime overhead. Related to NFR.S.4. Satisfies hazard H1-3 from the Hazard Analysis.
- **FR.S.2.2** The system shall report failure in the case that no valid partition across the cluster exists (e.g. not enough GPU-accessible memory).
- **FR.S.2.3** The system shall produce partitions that when assembled, are equivalent to the original model. Satisfies NFR.S.6.
- **FR.S.2.4** The system shall assign larger partitions to more capable devices[4]. Related to NFR.S.4.

S.2.3 Inference Engine

Functional Requirements

- **FR.S.3.1** The system shall load the model partition given by the local Coordinator into GPU-accessible memory.
- **FR.S.3.2** The system shall accept incoming tensors from Tensor I/O, and run them through the loaded partition.
- **FR.S.3.3** The system shall send output tensors to Tensor I/O.
- **FR.S.3.4** The system shall measure partition performance metrics[5]

[4]The "capabilities" of any given devices are defined via a combination of network throughput, GPU TFLOPs/TOPs, GPU-accessible memory size, and GPU-accessible memory bandwidth (GB/s).

[5]e.g. token throughput, end-to-end partition latency, backpressure.

and provide them to the local Coordinator on a 5 second interval.

- **FR.S.3.5** The system shall support processing tensors from multiple different requests concurrently. Related to NFR.S.5.

S.2.4 Tensor I/O

Functional Requirements

- **FR.S.4.1** The system shall establish connections with neighbouring nodes' Tensor I/O components.
- **FR.S.4.2** The system shall dynamically choose the compression level of outgoing tensors to maximize throughput based on available CPU-time and network throughput.
- **FR.S.4.3** The system shall accept input tensors from the upstream node, decompress them, and provide them to the Inference Engine.
- **FR.S.4.4** The system shall accept output tensors from the Inference Engine, compress them, and send them to the downstream node's Tensor I/O.
- **FR.S.4.5** The system shall measure network performance data[6], for established connections, and provide that data to the local Coordinator on a 5 second interval.
- **FR.S.4.6** The system shall maintain request association when sending tensors[7]. Related to NFR.S.5.

S.2.5 Web UI

Functional Requirements

- **FR.S.5.1** The system shall allow a user to scan for other connected devices running Lake. Partially satisfies NFR.S.2.
- **FR.S.5.2** The system shall allow a user to select connected devices to create a cluster. Partially satisfies NFR.S.2.

[6] e.g. RTT, throughput, backpressure.
[7] This is to allow tensors from different concurrent requests to be differentiated.

- **FR.S.5.3** The system shall allow a user to select an input GGUF file. Partially satisfies NFR.S.2.
- **FR.S.5.4** The system shall display an error to the user if partitioning the selected input file is infeasible given the selected devices. Partially satisfies NFR.S.1.
- **FR.S.5.5** The system shall display the performance metrics of the cluster as a whole, as well as for each individual node and connection. Partially satisfies NFR.S.1.
- **FR.S.5.6** The system shall highlight nodes or network connections causing bottlenecks (TODO: glossary bottleneck) in the cluster. Partially satisfies NFR.S.1. Satisfies hazard H4-3 from the Hazard Analysis.
- **FR.S.5.7** The system shall, upon receiving an invite to join a cluster, display a prompt for the user to either confirm or decline the invitation. Satisfies NFR.S.12.

S.2.6 Non-Functional Requirements

S.2.6.1 Usability

- **NFR.S.1 Understandability:** The user must be able to easily understand cluster information. A new user must be able to describe which devices are participating, the token throughput of the system, and which devices/connections are bottlenecks within 2 minutes of seeing the interface. Partially satisfied by FR.S.5.4, FR.S.5.5, and FR.S.5.6.
- **NFR.S.2 Controllability:** The user must be able to easily control cluster configuration. A new user must be able to create a new cluster within 2 minutes of seeing the interface. Partially satisfied by FR.S.5.1, FR.S.5.2, and FR.S.5.3.
- **NFR.S.3 Installability:** The user must be able to easily install Lake natively on Windows, macOS, and Linux. On Windows and macOS, the installation process should be automated and should require at most 10 clicks after downloading the installer. On Linux, the installation process should only require a single package manager command.

S.2.6.2 Performance

- **NFR.S.4 Model Size:** The cluster must be able to run a model with more parameters, higher precision, or larger context window than would be able to run individually on the node with the most memory. Related to FR.S.2.1, and FR.S.2.4.
- **NFR.S.5 Scalability:** The number of concurrent requests with ¡ 20% individual request performance degradation shall scale linearly with the number of participating devices. Related to FR.S.1.14, FR.S.3.5, and FR.S.4.6.
- **NFR.S.6 Intelligence:** The partitioned model must score at least 95% as well as the same model running on one machine using relevant benchmarks[8]. Satisfied by FR.S.2.3.

S.2.6.3 Look and Feel

- **NFR.S.7 Consistency:** Colors and fonts must be used consistently across the application.

S.2.6.4 Fault-Tolerance

- **NFR.S.8** The system shall reject invalid or corrupted GGUF files, and notify the user of the issue.
- **NFR.S.9** The system shall produce correct results in the face of network packet corruption and dropping up to 10% packet loss. Partially satisfies hazard H3-1 from the Hazard Analysis.

S.2.6.5 Privacy

- **NFR.S.10** The system shall not save any request data[9] after the request has completed.

[8]Relevant benchmarks are entirely dependent on what types of model are being tested. Example benchmarks: SWE-bench, FrontierMath

[9]**Request data** refers to any data sent to Lake as part of the request, or any data that would give insight into what the data sent in the request might have been.

- **NFR.S.11** The system shall not send **any** data to any device other than those included in the cluster. This includes, but is not limited to all kinds of request data, telemetery, and user data.
- **NFR.S.12** The system shall not include untrusted devices in a cluster. Satisfied by FR.S.1.17, FR.S.1.18, FR.S.1.20, FR.S.1.21, and FR.S.5.7. Satisfies hazard H5-1 from the Hazard Analysis.

S.2.6.6 Compatibility

- **NFR.S.13 Operating System:** The system shall be compatible with Windows, macOS, Linux.
- **NFR.S.14 CPU Architecture:** The system shall be compatible with x86_64 and ARM CPU architectures.
- **NFR.S.15 GPU Technology:** The system shall be compatible with NVIDIA, AMD, and Apple GPUs.

S.3 Interfaces

Lake requires the following interfaces:

- **OpenAI compatible API** – An HTTP server, following the OpenAI API structure. This will be the API used to interact with the user, via third-party tools of the user's choice that accept the OpenAI API (e.g. OpenCode).
- **GGUF input model** – An input file using the GGUF file format. This contains the model Lake will partition and run.
- **Rust-to-C FFI** – The Rust-to-C foreign function interface will be used to call ggml from the Inference Engine.
- **gRPC between nodes** – A gRPC-based API used to communicate between nodes.
- **REST API between Coordinator and Web UI** – A REST API for communication between the leader Coordinator and the Web UI.

S.4 Normal Operation

1. User opens Lake instances on all participating machines.
2. User creates a Lake cluster using the UI on one of the devices, and provides a GGUF file. This device becomes the leader node.
3. Leader node checks connectivity with follower nodes, and collects performance metrics for each node.
4. Partitioner devises a partitioning scheme based on the collected metrics.
5. Leader node sends the model partitions to their respective devices.
6. Neighbouring nodes on the partition graph establish direct P2P connections.
7. User initiates inference request by providing a prompt via the OpenAI compatible API.
8. Leader node sends the prompt tokens to the inference engine corresponding to the first node of the graph, initiating prefill.
9. Prompt tokens propagate through the graph until prefill completes, and the first output token is generated.
10. Output tokens are repeatedly fed back into the top of the graph, generating a new token on each iteration until the response is complete.
11. As the response generates, the user receives the generated output via the OpenAI compatible API.
12. User initiates a shutdown of the Lake cluster.
13. Each node unloads its portion of the graph.
14. Leader node saves the cluster configuration to disk.
15. All nodes sever their connections, and all Lake instances terminate.

S.5 Undesired Event Handling

- **Model is too large for the available memory in the cluster** – The Partitioner reports a partitioning failure, and this error is communicated to the user via the Web UI.
- **Follower node fails during inference** – Leader node re-tries inference, if it continues to fail, the leader rebuilds the cluster, excluding the problematic device. Associated with hazard H2-1 in the Hazard Analysis.
- **Tensors are corrupted on transfer between nodes** – The receiving Tensor I/O requests that the tensor is retransmitted from the source Tensor I/O. Associated with hazard H3-1 in the Hazard Analysis.
- **Connection between two nodes is lost** – The nodes in question attempt to re-establish a connection. If this fails, the leader must rebuild the cluster, excluding the failed link. Associated with hazard H4-1 in the Hazard Analysis.
- **Input GGUF file is corrupted or fails to load** – Refuse to begin inference, and inform the user of the corrupted or faulty file.
- **A request is made with a larger prompt than the cluster has memory for** – If there are existing requests running, wait for them to complete, then try again. If there is still not enough memory, return an error for the request, and instruct the user to shorten their prompt or context window.
- **A rebuilt cluster is too small to hold the model after a node leaves** – The Partitioner reports a partitioning failure, and this error is communicated to the user via the Web UI.
- **A user attempts to create a cluster with a node without a GPU** – The Web UI displays an error message outlining the requirement for all nodes to have a GPU.

S.6 Detailed Usage Scenarios

[Full use cases for the scenarios introduced in G.5, one per scenario, in the structure below. The main success scenario should be numbered so extensions

can reference its steps. —TPLT]
Use Case Name: Name
Primary Actor: Actor
Stakeholders and Interests:

- Stakeholder: interest.

Preconditions:

- Precondition.

Postconditions (Success Guarantees):

- Postcondition.

Postconditions (Failure Guarantees):

- Postcondition.

Main Success Scenario (Basic Flow):

1. Step.

Extensions:

- **Na.** Alternative or failure at step N, and how it is handled.

Justification: Why this scenario matters.

S.7 Prioritization

[Every requirement from S.2 and the Environmental Requirements table, with a MoSCoW priority and the reason for it. Group the tables by component. The Date column records when the priority was last decided. —TPLT]
Priorities follow MoSCoW: **M** must have, **S** should have, **C** could have, **W** won't have (this release).

Component Name

ID	Requirement Name	Priority	Reasoning	Date
FR.S.1.1	Name	M	Reasoning	YYYY-MM-DD
FR.S.1.2	Name	S	Reasoning	YYYY-MM-DD

Table 8: Prioritization of Component Name requirements

S.8 Verification and Acceptance Criteria

[How each class of requirement will be shown to be satisfied: unit and integration tests, measurements, user studies, inspections. Make the criteria concrete enough that the V&V Plan can be derived from them. Every requirement in S.2 should map to at least one criterion here. —TPLT]

Validation Overview:

Verification Approach by Requirement Class:

- **Class of requirements** – How it will be verified and what counts as acceptance.

S.9 Formalizations

Two parts of Lake are formalized. S.9.2 defines what counts as a valid partitioning scheme for the model across the cluster, which every other requirement depends on. S.9.3 specifies, as a finite state machine, how the system responds when nodes join and leave the cluster. This is the precise form of the Dynamic Allocation goal.

S.9.1 Notation

For integers a and b, $a..b = \{\, n \in \mathbb{N} \mid a \leqslant n \leqslant b \,\}$, which is empty when $a > b$.

Table 9: Symbols used in the formalization

Symbol	Type (unit)	Meaning
N	$\mathbb{N}$, $N \geqslant 1$	Number of partition units in the model (embedding, transformer layers and output head), in execution order.
L	$L = 1..N$	The set of partition units.
D	finite set	The nodes currently in the cluster: devices the user has selected (FR.S.5.2) and that have confirmed participation (FR.S.1.21).
m_ℓ	$\mathbb{N}$ (bytes)	Memory needed to hold unit ℓ's weights and its KV cache for $CONTEXT_MAX$ tokens.
M_d	$\mathbb{N}$ (bytes)	GPU-accessible memory free on node d when it joins the cluster, minus the runtime overhead reserved by FR.S.2.1.

The formalization assumes that M_d does not change while d is in the cluster, and that $join(d)$ only occurs for $d \notin D$ and $leave(d)$ only for $d \in D$.

S.9.2 Valid Partitioning Schemes

Definition S.9.2.1 (Partitioning scheme)**.** A *partitioning scheme* is a finite sequence of partitions

$$P = \langle (d_1, a_1, b_1), \ldots, (d_k, a_k, b_k) \rangle, \qquad k \geqslant 1,$$

where d_i is a node and $a_i, b_i \in \mathbb{N}$. Partition i consists of units $a_i..b_i$, runs on node d_i, and passes its output to partition $i + 1$. The nodes of P are $dev(P) = \{d_1, \ldots, d_k\}$.

Definition S.9.2.2 (Valid partitioning scheme)**.** P is *valid* over cluster D, written $P \in \mathcal{V}(D)$, if and only if:

(V1) $dev(P) \subseteq D$ and $d_i \neq d_j$ for all $1 \leqslant i < j \leqslant k$;

(V2) $a_1 = 1$ and $b_k = N$;

(V3) $a_i \leqslant b_i$ for all $i \in 1..k$;

(V4) $a_{i+1} = b_i + 1$ for all $i \in 1..k - 1$;

(V5) $\sum_{\ell=a_i}^{b_i} m_\ell \leqslant M_{d_i}$ for all $i \in 1..k$.

(V1) gives each node at most one partition. (V2)–(V4) make the partitions consecutive, non-empty blocks of units in execution order. (V5) says each node has room for its partition. The Model Partitioner must return a member of $\mathcal{V}(D)$ whenever $\mathcal{V}(D) \neq \emptyset$, and must report failure otherwise (FR.S.2.2).

Lemma S.9.2.3 (Exact cover)**.** *If $P \in \mathcal{V}(D)$, every unit $\ell \in L$ belongs to exactly one partition. That is, the sets $a_i..b_i$ for $i \in 1..k$ are non-empty, pairwise disjoint, and their union is L.*

Proof. Let $b_0 = 0$. By (V2) and (V4), $a_i = b_{i-1} + 1$ for all $i \in 1..k$, so partition i consists of units $(b_{i-1}+1)..b_i$. By (V3) these sets are non-empty, so $0 = b_0 < b_1 < \cdots < b_k = N$. Each $\ell \in L$ therefore satisfies $b_{i-1} < \ell \leqslant b_i$ for exactly one i, so it belongs to exactly one partition. □

S.9.3 Allocation State Machine

The system's response to nodes joining and leaving the cluster is a finite state machine (Q, Σ, δ, q_0) whose transitions are guarded by predicates over the cluster D and the active partitioning scheme P.

S.9.3.1 States.

$Q = \{\mathsf{U}, \mathsf{A}, \mathsf{I}, \mathsf{G}, \mathsf{R}, \mathsf{X}\}$ with $q_0 = \mathsf{U}$:

- U (Unloaded): no model is loaded.
- A (Allocating): a target scheme P' is being loaded, and no request is in progress.
- I (Idle): the model is loaded on scheme P.
- G (Generating): a request is being served on P.
- R (Recovering): a node in $dev(P)$ left during a request. The request is stopped while P' is loaded, then retried (FR.S.1.13).
- X (Insufficient): $\mathcal{V}(D) = \emptyset$, so the cluster cannot hold the model.

S.9.3.2 Events. $\Sigma = \{load, unload, req, fin, done\} \cup \{join(d), leave(d)\}$:

- *load* and *unload*: the user creates a cluster with a model, or shuts it down.
- *req* and fin: a request is submitted, or finishes.
- *done*: the system finishes loading P'.
- $join(d)$: a device the user selected confirms participation and is added to the cluster.
- $leave(d)$: follower d leaves, or is detected as failing (FR.S.1.10).

This finite state machine only models the behaviour of follower nodes. If the leader node is detected as failed (FR.S.1.24), the followers shut down and the session ends (FR.S.1.25), which has the same effect as *unload*.

S.9.3.3 Guards. Let D' be the cluster after the event: $D \cup \{d\}$ for $join(d)$, $D \setminus \{d\}$ for $leave(d)$, and D otherwise. Then

$$\varphi \equiv \mathcal{V}(D') \neq \emptyset \qquad \text{and} \qquad \psi \equiv d \in dev(P).$$

Whenever the machine enters A or R, it selects a new target $P' \in \mathcal{V}(D')$.

S.9.3.4 Transitions. Table 10 defines δ. Combinations it does not list are handled as follows:

- *unload* moves every state to U and aborts any request in progress.
- $leave(d)$ with $\neg\psi$ in I or G removes d from D and leaves the state unchanged.
- *req* is rejected with an error in U and X. *req* in A, G or R, and $join(d)$ in G or R, are queued and fire in arrival order once I is entered. The machine therefore serves one request at a time, and concurrent requests (FR.S.1.14) are not modelled.
- Any other combination leaves the state unchanged.

Table 10: Transition function δ of the allocation state machine

#	State	Event [guard]	Next	Effect
1	U	*load* $[\varphi]$	A	Start loading P'.
2	U	*load* $[\neg\varphi]$	X	Report a partitioning failure.
3	A	*done*	I	$P := P'$.
4	A	$join(d)$, $leave(d)$ $[\varphi]$	A	Reselect P' and restart loading.
5	A	$join(d)$, $leave(d)$ $[\neg\varphi]$	X	Report a partitioning failure.
6	I	*req*	G	Start generating.
7	I	$join(d)$	A	Rebuild the cluster over D'.
8	I	$leave(d)$ $[\psi \wedge \varphi]$	A	Rebuild the cluster over D'.
9	I	$leave(d)$ $[\psi \wedge \neg\varphi]$	X	Report a partitioning failure.
10	G	*fin*	I	Return the response.
11	G	$leave(d)$ $[\psi \wedge \varphi]$	R	Stop the request.
12	G	$leave(d)$ $[\psi \wedge \neg\varphi]$	X	Abort the request with an error.
13	R	*done*	G	$P := P'$, retry the request.
14	R	$leave(d)$ $[\varphi]$	R	Reselect P' and restart loading.
15	R	$leave(d)$ $[\neg\varphi]$	X	Abort the request with an error.
16	X	$join(d)$ $[\varphi]$	A	Start loading P'.

Row 7 needs no guard. Conditions (V1)–(V5) refer to D only through $dev(P) \subseteq D$, so a scheme valid over D stays valid over $D \cup \{d\}$. For the same reason, removing a node outside $dev(P)$ keeps P valid.

S.9.3.5 Invariant. In states I and G, $P \in \mathcal{V}(D)$. P is only assigned in rows 3 and 13, from a P' that is reselected after every change to the cluster. The only change that keeps the machine in I or G removes a node outside $dev(P)$.

S.10 Likely Changes to Requirements

- **LC.1** A Lake cluster cannot recover from a failing leader node (see FR.S.1.25). In the future, the ability to recover from a failing leader node may be added.
- **LC.2** Lake only supports desktop operating systems (Windows, macOS, Linux) (see NFR.S.13). Support for mobile devices (iOS, Android)

may be added once other goals and requirements have been met.

- **LC.3** Lake assumes the local area network being used to connect the devices is private and trusted (See Hazard Analysis Section 4 - Critical Assumptions). Lake's may be extended with additional security hardening to allow clusters to run on public or untrusted networks.
- **LC.4** Lake requires that each device manually confirm joining a cluster (see FR.S.1.17 and FR.S.1.21). In the future, a trust mechanism may be introduced, which would allow devices to automatically join clusters created by trusted devices without requiring confirmation.
- **LC.5** Requirement thresholds and intervals will likely change, as real-world usage gives better estimates for reasonable threshold values.
- **LC.6** It is likely that future versions of Lake will support a headless mode where all UI interactions are performed through a command line interface. This would be useful for users wishing to use headless servers as nodes.

S.11 Unlikely Changes to Requirements

- **UC.1** Model training and fine-tuning support is unlikely to be added to Lake. Lake is primarily an inference framework, so the addition of training would be out of scope for such a project.
- **UC.2** Additional input model formats aside from GGUF (TODO: glossary) are unlikely to be added, as GGUF is one of the most common standards for LLMs (TODO: glossary). Additionally, models in other formats can be converted to GGUF so there is little reason for Lake to support them directly.
- **UC.3** A primary goal of Lake is privacy (see NFR.S.11), so modifications that would compromise privacy like incorporating cloud-based services are unlikely.
- **UC.4** Even though one of Lake's goals is to run larger models, integrating quantization tools into Lake is unlikely. Lake's scope is focused on distributed inference, and quantization tools would better fit in the scope of an independent project.

(P) Project

P.1 Roles and Personnel

- **Liaison (Ethan Walsh)** – Coordinates communication between the team and other project stakeholders and faculty.
- **Repository administrator (Matthew Nesbitt)** – Manages repository and organizational configurations.
- **Architecture lead (Ethan Walsh)** – Leads the architecture design of the system.
- **Coordinator lead (Gary Qin)** – Leads the coordinator and IPC design of the system.
- **Performance lead (Matthew Nesbitt)** – Leads the performance benchmarking and optimization of the system.
- **Machine learning leads (Jinwoo Hong, Colin Chambachan)** – Lead the ML systems of Lake.
- **Meeting chair (Rotating weekly amongst team members)** – Leads weekly team meetings.
- **QA lead (Rotating weekly amongst team members)** – Performs quality assurance and dogfooding of the system.
- **Teaching assistant (Rebecca Di Filippo)** – Provides technical guidance and feedback throughout the project.

P.2 Imposed Technical Choices

- **API specification** – Lake will follow the OpenAI API specification for its request and response formats (**?**). This is chosen due to the widespread adoption of this API specification and the flexibility it provides when connecting to existing external clients.
- **Supported platforms** – Windows, macOS, and Linux will be supported platforms for nodes within the Lake network. Android and iOS support is a stretch goal. These are chosen due to the availability of

inference engine support and hardware acceleration support for these operating systems.

P.3 Schedule and Milestones

Milestone	Due Date
Team formation and project selection	Sep 18, 2026
Problem statement, PoC plan, development plan, team charter	Sep 28, 2026
SRS and Hazard Analysis Rev 0	Oct 19, 2026
V&V Plan Rev 0	Nov 2, 2026
Design Document Rev -1 and CD-Fall	Nov 16, 2026
PoC demonstration	Nov 23–Dec 3, 2026
Design Document Rev 0	Jan 18, 2027
Peer module implementation	Jan 21, 2027
Rev 0 demonstration	Feb 1–11, 2027
V&V Report and CD-Winter	Mar 8, 2027
Final demonstration	Mar 15–21, 2027
Final documentation	Mar 31, 2027
EXPO poster and video	Mar 31, 2027 (tentative)
EXPO demonstration	April 6, 2027 (tentative)

Table 11: Project Milestones and Due Dates

P.4 Tasks and Deliverables

[The work breakdown: each task, what it produces, and which milestone it feeds. Tasks should be small enough to assign to one or two people. —TPLT]

Task	Description / Steps	Deliverable	Due Date
Task	Description	Deliverable	Date

Table 12: Tasks and Deliverables

P.5 Required Technology Elements

[Everything the project needs to obtain or build before the system can be delivered: libraries, hardware, datasets, services, licences, compute. Note anything not yet secured. —TPLT]

- **Element** – Purpose and status.

P.6 Risk and Mitigation Analysis

[Project risks (schedule, technical, people, external dependencies), each with its likelihood, impact, and mitigation. System hazards belong in the Hazard Analysis, not here. —TPLT]

- **Risk: Name**
  - Description:
  - Mitigation:

P.7 Requirements Process and Report

[How requirements were elicited and how they will be maintained: who was consulted, through which channels, how changes are tracked, and how this document will be kept in sync with the Hazard Analysis, V&V Plan and design. —TPLT]

- **Elicitation** –
- **Communication** –
- **Change Tracking** –
- **Process Updates** –

Safety and Security Requirements

Functional Safety Requirements

- **SR0-1** – The system shall identify and discard stale device health data from determining device availability by configuring a time-to-live for valid data.
- **SR0-2** – The system shall only route inference requests to nodes that currently have a confirmed model layer partition.
- **SR1-1** – The system shall assign at least two nodes for model layer partitioning during distributed inference.
- **SR1-2** – The system shall ensure all model layers are assigned to a node, and that each device configured to run model layers receives a non-zero model layer partition assignment.
- **SR1-3** – The system shall reject models and layer allocations that are estimated to exceed the memory capacities of the Lake network or individual nodes, including a configured safety margin. Related SRS requirements: FR.S.2.1, FR.S.2.2.
- **SR2-1** – The system shall attempt an automatic bounded recovery and retry on inference requests if a follower node fails during inference. If the follower node is unable to be recovered within the configured recovery limits, the system must report to the API that the request has failed. Related SRS requirements: FR.S.1.10, FR.S.1.11, FR.S.1.13.
- **SR3-1** – The system shall identify corrupted and incomplete tensors throughout the distributed inference with integrity checks and communication timeouts to discard affected data and retry requests.
- **SR3-2** – The system shall log failed tensor decoding metadata and report a failure to the coordinator so the inference request can recover, subject to SR5-2.
- **SR4-1** – The system shall configure a timeout for communication between nodes, and handle exceeded timeouts according to the appropriate recovery or failure policy.

- **SR4-2** – The system shall mark a node as unavailable for inference or model layer partitioning when the node health criteria are not met, and require a successful health check before marking the same node as available. During a leader node failure, follower nodes shall detect and initiate a shutdown. Related SRS requirements: FR.S.1.24 and FR.S.1.25.

- **SR4-3** – The system shall utilize configured values or historical data to determine a baseline inter-node communication latency and notify the coordinator when that baseline is exceeded by a specified tolerance.

- **SR5-1** – The system shall require user approval on the leader node to authorize a follower node, and user confirmation on the follower node to confirm the authorization before it can participate in the Lake network. Related SRS requirements: FR.S.1.17, FR.S.1.18, FR.S.1.20, FR.S.1.21, NFR.S.12.

- **SR5-2** – The system shall not record prompts, responses, tensor data, and KV cache content in diagnostic logs. Related SRS requirements: NFR.S.10.

Requirement Dependencies

[A traceability diagram or matrix showing which requirements depend on which. Use the landscape environment below for a wide figure. —TPLT]

[Cite sources with \citep{key} or \citet{key}; entries live in refs/References.bib. The References section appears once at least one source is cited. —TPLT]

1 Appendix — User Interview

The questions below were sent, on October 16, 2026, to an AI developer at an aerospace engineering firm who runs the firm's in-house LLM assistant on its own hardware, and he answered in writing. The answers are summarised; the full notes are kept with the team's records. How they shaped the Goals chapter is described in Section G.7.2.

1. *What do you use to run models locally today, and on what machine?* vLLM for the main models, Ollama for small vision models, and a text-embeddings server, all in Docker on Linux, on two GPU workstations with 96 GB and 48 GB cards. The main model is a 27-billion parameter model at full precision with a 131k-token context.

2. *What stops you from running a bigger model?* Memory per card: the 27B model fills most of a 96 GB card once conversation context is added, and a 70B-class model would need both cards or compression. Also prompt-reading time on long agent prompts, and leaving room for concurrent users. Model quality is already good enough for their tasks.

3. *Do you have idle computers, and would you pool them?* Most office machines are laptops with integrated graphics; some GPU workstations sit idle at night. Yes for background jobs (re-indexing, OCR, embeddings, overnight test runs); probably not for live chat, since a split over the office network would likely be slower than one good GPU. Conditions: data control on every machine, backing off instantly when the owner is working, tolerance for machines sleeping or updating, network speed, and IT approval with signed installers and an off switch.

4. *If responses were slow, what would you want to see?* A per-request timeline per device with the slowest highlighted, per-link bandwidth and latency, a comparison against the single best machine, whether the prompt cache was hit, and metrics on by default and exportable.

5. *If a machine dropped out mid-answer?* Retry quietly once, continuing from where it stopped; if that fails, tell the user and show what was written so far. Never return a cut-off answer that looks complete, and never repeat an action, since their agent drafts emails and edits files. Log every drop-out and remove machines that keep failing.

6. *How much does keeping data on your network matter?* It is the reason they run locally: cloud models cannot legally process export-controlled documents. Anything like Lake must run fully offline with no telemetry, encrypt traffic between nodes, accept only company-managed machines, and be auditable. Shared or public networks: no.

7. *Which tools would you use it from?* An OpenAI-compatible API is a must, with streaming, tool calling, strict JSON output, image input, and options such as turning "thinking" off. Then a chat window, a coding assistant for code that cannot go to the cloud, and batch scripts.

8. *Command line or installer and UI?* Command line and Docker on servers, with configuration in files and pinned versions. On colleagues' machines, a signed installer pushed by IT, running as a background service with a pause switch. A small admin page for machine health and timing. No automatic updates.

2 Appendix — Generative AI Use Disclosure

[ChatGPT (version 5) was used to brainstorm ideas for the template for this disclose section. —SS]

2.1 AI Tools Used

[Give the name of the AI tool(s) used and the version. —SS]

2.2 How and Where Used

[Explain what the tool(s) were used for. Examples include: brainstorming, explaining a technical concept, reviewing the document, generating or modifying text, generating or modifying diagrams, identifying errors or omissions or inconsistencies. —SS]

[For each of the uses, identify which parts of the document were affected. —SS]

[You do not need to report every interaction, but you should disclose the uses that materially contributed to your work —SS]

2.3 Verification of AI-Generated Content

[

- Describe how the team checked AI-generated or AI-assisted material before including it in the document.
- In particular, verify factual claims, technical assertions, requirements, calculations, references, and other information for which an AI system may produce incorrect or fabricated results.
- **The team is responsible for the accuracy, completeness, consistency, and appropriateness of the submitted document, regardless of whether material was generated by a human or an AI system.**

—SS]

3 Appendix — Reflection

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?

2. What pain points did you experience during this deliverable, and how did you resolve them?

3. How many of your requirements were inspired by speaking to your client(s) or their proxies (e.g. your peers, stakeholders, potential users)?

4. Which of the courses you have taken, or are currently taking, will help your team to be successful with your capstone project.

5. What knowledge and skills will the team collectively need to acquire to successfully complete this capstone project? Examples of possible knowledge to acquire include domain specific knowledge from the domain of your application, or software engineering knowledge, mechatronics knowledge or computer science knowledge. Skills may be related to technology, or writing, or presentation, or team management, etc. You should look to identify at least one item for each team member.

6. For each of the knowledge areas and skills identified in the previous question, what are at least two approaches to acquiring the knowledge or mastering the skill? Of the identified approaches, which will each team member pursue, and why did they make this choice?